CAMARINES SUR POLYTECHNIC COLLEGES
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CAMARINES SUR POLYTECHNIC COLLEGES
COLLEGE of COMPUTER STUDIES

SYNTIX: AUTOMATED INTRAMURALS MULTI-EVENT SCORING AND TALLYING SYSTEM FOR CSPC SUPREME STUDENT COUNCIL

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By

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CHAPTER 1
INTRODUCTION

The use of digital technology is to maintain the efficiency and quality of institutional operations is an imperative issue in contemporary on higher education. More and more educational institutions are changing from manual labor to automated solutions driven by real time data processing and information feedback. The change is important for big event management because the accuracy of results and the timely delivery of results directly impact the student body's engagement and trust.

Camarines Sur Polytechnic Colleges (CSPC) as an ISO 9001:2015 Certified institution continuously pursues excellence in administrative and other academic and extracurricular services. One of the most important events of the institution is the Intramurals. This multi-event competition includes different sports, literary-musical contests and pageants. Because of the reliance on manual paper-based tallying, there has been a history of system breakdowns such as delays in announcements and mathematical errors that affect the spirit of the games.

The Automated Intramurals Multi-Event Scoring and Tallying System (SYNTIX) is a web-based system that will automate the scoring of events of CSPC Supreme Student Council. The project has been given a particular context. Also, has various specific objectives to solve already existing inefficiencies. Furthermore, the importance of the system to the stakeholders. Also, the scope and limitations of the system.

1.1 Project Context

Camarines Sur Polytechnic Colleges (CSPC) academic community Certified with ISO 9001:2015, has its benefits for students' development when participating in Intramural events. However, student involvement is affected by poor event management. Currently, the Supreme Student Council (SSC) is using a strategy that will lead to exposure of hasty generalizations, lagged clerical processes, and operational backlog. Gula (2023) mentioned that the absence of prizeallocations in a sporting event is a consequence of poor result management. Cualbar (2024) argued that the absence of digitization in the score tallying process, which is one of the most important elements for the success of the event, is one of the most important criteria that should be taken into account when trying to improve speed and accuracy in order to eliminate the score tallying process.

To tackle these issues, SYNTIX is created as a sturdy, web-based platform that can manage intricate outcomes from numerous sporting and cultural events in real-time. As noted by Ramizares (2022) [8], adaptive multi-event systems are crucial for updating school competitions, a vision supported by the design of real-time scoring engines mentioned by Thompson (2024) [9]. By automating the calculations and tallying processes, the system promotes trust and transparency within the student community through the prompt distribution of precise results [1] Al-Zubidi. (2023). This digital shift guarantees that the institution stays at the leading edge of technological integration in campus activities.

Aside from the aspect of technical efficiency, the development of SYNTIX also considers quality standards and legal compliance to conform to the standards of an ISO certified institution. The user interface of the system is based on the usability standards set in the ISO 9241-11:2018 [4] and the design heuristics proposed by Nielsen (2025) [6], considering that the system should be user-friendly for the judges and organizers to use. Furthermore, the development of the system is committed to the standards set in the Republic Act No. 10173, or the Data Privacy Act of 2012 [7], considering the confidentiality of the students' information, as emphasized in the quality management principles that significantly contribute to the performance of the university, "substituting the traditional manual constraints with a reliable and secure solution for the CSPC community," as mentioned in the study of [5] Iqbal. (2025).

1.2 Purpose and Description of the Project

The purpose of SYNTIX is to improve the efficiency, accuracy, and dependability of the management of multi-event competitions during the Intramurals at Camarines Sur Polytechnic Colleges (CSPC). The project focuses on solving the problems of the manual calculation of scores and the lack of transparency of results, which causes confusion to the student-athletes and student performers.

The study optimistically aims to minimize clerical fatigue of the Supreme Student Council (SSC) and the board of judges and high data integrity due to the absence of human mathematical errors. The system foremost aims to inspire confidence in competition results and secondary aims to provide a professional, easy, and inclusive system to record and announce scores on demand.

SYNTIX is a web-based application built using the Laravel framework, allowing judges, organizers, and students to communicate through a centralized server architecture. The unique functionality of SYNTIX includes an automated tallying feature that incorporates a variety of scoring systems, from point-based to percentage-based systems, managed through a secured MySQL relational database. In addition, judges can enter their scores via a judge scoring interface, which eliminates the use of paper scoring sheets and the need to transport the sheets. Students can view the updated rankings via a dynamic leaderboard powered by RESTful APIs, and tabulators can record and keep track of sports results right in the system. Only authorized administrators can access the secure Administrative Dashboard, which lets them keep track of and manage the counting of several events at once in real time.

What makes SYNTIX unique is its adaptable, multi-event architecture and its focus on institutional transparency for a CSPC environment. Unlike typical spreadsheets or simple tabulation tools, SYNTIX contains a "double-entry" prevention mechanism and automated validation integrated into its backend logic that will not accept anything over the pre-set criteria for precision during extreme pressure conditions. SYNTIX also has the unique ability to provide instant, official electronic summaries and results while strictly adhering to the Data Privacy Act (RA 10173) and ISO standards for end-user usability, offering unparalleled security and professionalism as compared to manual systems and generic applications. By combining server-side data processing with strong protection, SYNTIX serves as a modern and secure solution that meets the current sports management standards of colleges.

1.3 Objectives of the Project

This study aims to design and develop SYNTIX: Automated Intramurals Multi-Event Scoring and Tallying System, a web-based platform that will streamline the management of multi- event competitions for the CSPC Supreme Student Council. It serves as a centralized platform that facilitates efficient real-time score entry and automated result dissemination for the student body, while simultaneously automating mathematical tallies and record management to allow organizers and judges to focus more on the integrity of the games.

Specifically, this study aims to:

1. To identify the problems encountered by the CSPC – Supreme Student Council Managing the following Process:

   1.1 Manual collection and hand-encoding of physical score sheets.

   1.2 Delayed result announcements.

   1.3 Mathematical computation errors.

   1.4 Lack of real-time transparency for students.

   1.5 Inefficient retrieval of historical records.

2. To develop SYNTIX as a PWA-based Automated Tabulation and Event Management system With the following functional modules:

   2.1 User Management (Judges, Admin, and Students).

   2.2 Real-time Digital Scoring Interface.

   2.3 Automated Tabulation and Ranking Engine.

   2.4 Live Leaderboard and Results Dashboard.

   2.5 Local Data Persistence and Synchronization for offline-ready score entry (PWA).

   2.6 Historical Data Archive and Reports Generation

3 To evaluate the system using black box testing in terms of:

   3.1. Functionality;

   3.2. UI Testing;

   3.3. Security Testing; and

   3.4. User Acceptability;

1.4 Significance of the Study

The results of this study were deemed significant to the following.

Camarines Sur Polytechnic Colleges (CSPC). CSPC stands to benefit from the modernization of its event management systems. **SYNTIX** aligns with the institution's mission to deliver innovative and student-centered services. By integrating digital solutions into its activities, CSPC demonstrates its commitment to enhancing the student experience and maintaining high standards of data integrity as an ISO 9001:2015 Certified institution.

CSPC Supreme Student Council (SSC). SYNTIX offers significant improvements to the student council by streamlining the administrative side of the event. Observations of previous intramurals reveal that manual setups result in computation errors, misplaced score sheets, and heavy administrative burdens on student leaders. With this automated system, the SSC can manage multiple events simultaneously and focus more on quality coordination rather than tedious paperwork, ultimately enhancing the council's service delivery.

School Administrators. For administrators, the system presents a valuable monitoring and evaluation tool. It allows them to oversee how school-wide events are conducted and ensures that the results are evidence-based and free from mathematical disputes. By having access to real-time reports, administrators can uphold the institution's commitment to digital transformation and transparency.

Students. Students are the primary beneficiaries of **SYNTIX**. Through this platform, they gain faster and more convenient access to results without the need to wait for physical announcements. The availability of a live leaderboard empowers them to stay engaged, fosters healthy competition, and removes the frustration often felt when results are unclear. This is vital for maintaining high school spirit throughout the event.

Judges. The system provides a formal and efficient digital tool for the board of judges. By using a responsive web interface, judges can submit evaluations directly from their devices, eliminating the need for handwritten forms that are often prone to being lost or misread. This ensures that their professional judgment is processed accurately and immediately.

Researchers. This study is significant to the researchers as it serves as a practical application of the knowledge they have acquired in advanced web development and database management. Through this project, they can gain technical proficiency that will be essential in their future careers as Information Technology professionals.

Future Researchers. This research project can be considered a great source of information and base for future studies of those who seek out new ways to improve school administration systems. The results and design achieved during this project can be used as a basis for new projects focusing on real-time data analysis and counting algorithms. Further advancements are sure to come from future researchers building upon the results found here.

1.5 Scope and Limitations of the Project

This project focused on the development of the SYNTIX: Automated Intramurals Multi-Event Scoring and Tallying System for CSPC-Supreme Student Council, a web-based platform designed to streamline multi-event competitions through six core functional modules: a User Management Module providing Role-Based Access Control (RBAC) with specific interfaces for Judges, Tabulators, and Students; a Real-time Digital Scoring Interface for paperless, subjective scoring directly from devices; an Automated Tabulation and Ranking Engine for the instant computation of results based on event-specific criteria and weighted percentages; a Live Leaderboard and Results Dashboard for transparent real-time broadcasting and view-only access for the student body; a Local Data Persistence and Synchronization Module utilizing Progressive Web App (PWA) technology to allow score entry during signal fluctuations with automated transmission to a centralized Laravel backend and MySQL database; and a Historical Data Archive and Reports Generation Module for the authorized production of official PDF reports such as Detailed Judges' Score Breakdowns, Sports Result Summaries, and the Overall General Championship Medal Tally. The system features a secured Administration Dashboard that enables authorized SSC members to dynamically manage multiple events, including the creation of sports or cultural categories, encoding specific criteria, and the registration of teams and players.

Furthermore, the PWA architecture ensures an Offline Entry Capability, enabling judges and tabulators to continue data entry which will be processed and saved to the database as soon as a stable connection is established.

The system is limited to carry out the functions of scoring and tabulation, and it does not have auxiliary functions such as registration, tickets, budgeting, and video streaming. While the PWA provides the option for offline data entry as part of the contingency, it is, however, necessary to have a network connection to carry out the initial user login, real-time broadcasting to the student body, and final data synchronization to the centralized MySQL server. The access to the system is limited to authorized personnel, and the student interface is limited to "View-Only" leaderboard access to maintain data integrity. Last but not least, the system is intended to be a technical aid to eliminate mathematical errors and clerical fatigue, and it is not intended to replace the final decision-making authority vested in the Board of Judges and the Grievance Committee for subjective decisions and protests.

1.6 Project Dictionary

The following are the technical terms relevant to this study each defined both conceptually and operationally to ensure clear communication between the researchers and readers.

Real-Time Leaderboard. A web-based digital tool designed to provide live updates of standings to increase audience engagement and trust. This allows students to view live point aggregates, medal tallies, and rankings remotely via mobile or desktop devices to eliminate doubts regarding the fairness of the scoring process.

Black-Box Testing. A method of software testing that examines the functionality of an application without peering into its internal structure or code. This involves performing end-to-end tests, such as inputting a judge's score and verifying that the real-time leaderboard updates correctly, to ensure all system modules are fully functional.

SYNTIX. AWeb based automated Multi Event Scoring & Tabulation System which was specifically created to simplify the process of managing sports and cultural events by way of a digital platform. This pertains to the software that was created specifically for CSPC – Supreme Student Council, and makes use of the PWAs technology along with Laravel backend.

Background Synchronization. Generally, a web API that allows applications to defer tasks until the user has a stable internet connection. This is the automated process that detects when a judge's device regains internet access after being offline, triggering the system to "push" all locally cached scores to the central database without requiring manual user action.

Offline-First Architecture. A software design approach where an application is built to be fully functional even without an active internet connection by prioritizing local data storage. This means the scoring modules are designed to load and save data to the browser's internal memory (IndexedDB) during network outages, ensuring that the intramural event continues without data loss.

Progressive Web App (PWA). A type of application software delivered through the web, built using common web technologies intended to work on any platform that uses a standards- compliant browser. This refers to the system's responsive web design optimized for various devices, utilizing service workers to allow facilitators to access the platform even during network interruptions.

Role-Based Access Control (RBAC). An approach to restricting system access to authorized users based on their specific roles within an organization. This ensures that only authorized judges can access score input forms and that only administrators can manage event settings and finalize results.

UI Testing. The process of testing the interface of an application with the aim of ensuring that the user enjoys a seamless experience when using the software on various devices is referred to as the interface testing process. It ensures that the response from the SYNTIX dashboard on both mobile and desktop interfaces is satisfactory and that all buttons and forms are functioning.

Security Testing. This is a method of software evaluation which seeks to unearth any potential weaknesses and threats associated with an application, so as to secure any information against any form of tampering or unauthorized access. The validation of Role-Based Access Control (RBAC) and Laravel security systems will go a long way in ensuring that scores can be entered by authorized individuals but not viewed by the students.

User Acceptability Testing. The last stage of the software test life cycle in which the genuine end users test the software to check whether it can perform the necessary work based on their specifications in practical conditions. In this regard, a simulation stage is needed wherein the CSPC-SSC officers and judges will try out the program to verify whether its automatic tabulation, synchronization, and reporting modules can satisfy their needs.

Notes

[1] gustAl-Zubidi, A. 2023. Web-Based Competition Management System: A Tool for Transparent and Accurate Results. International Journal of Computing. Retrieved March 7, 2026 from https://ijc.org/web-competition-2023.

[2] Cualbar, J.L., Candia Jr., A.D., Abcede, J.M.L., and Genalazo, A.R.A. 2022. E-Tabulation System for Events and Activities with LCD Screen Integration. ResearchGate. Retrieved March 7, 2026 from https://www.researchgate.net/publication/360000000.

[3] Gula, L.P. 2023. Ineffective Practice and Lack of Motivation of the Junior High School Athletes for Physical Training in Preparation for City Meet and EVRAA. FDLA Journal. Retrieved March 7, 2026 from https://fdla.org/journal-2023.

[4] International Organization for Standardization. 2018. Ergonomics of human-system interaction — Part 11: Usability: Definitions and concepts (ISO Standard No. 9241-11:2018). Retrieved March 7, 2026 from https://www.iso.org/standard/63500.html.

[5] Iqbal, S., Taib, C.A.B., Allumi, N.A., and Abbas, M. 2025. Quality-driven education: Analyzing the effects of ISO 9001 quality management principles on university performance. Journal of Cultural Analysis. Retrieved March 7, 2026 from https://jcasc.com/quality-education-2025.

[6] Nielsen, Jakob. 2025. Heuristics for User Interface Design and User-Friendly Systems. Nielsen Norman Group. Retrieved March 7, 2026 from https://www.nngroup.com/articles/ten-usability- heuristics.

[7] Official Gazette of the Republic of the Philippines. 2012. Republic Act No. 10173: Data Privacy Act of 2012. Retrieved March 7, 2026 from https://www.officialgazette.gov.ph.

[8] Ramizares, M.L. 2022. Development of a Web-Based Adaptive Multi-Event Tabulation System.ResearchGate. Retrieved March 7, 2026 from

https://www.researchgate.net/publication/380000000.

[9] Thompson, Billie. 2024. The Architecture of Real-Time Scoring Engines in Web Applications. Journal of Web Engineering. Retrieved March 7, 2026 from https://example-journal.com/web- scoring-202

CHAPTER 2
REVIEW OF RELATED LITERATURE AND SYSTEMS

This chapter includes a comprehensive review of existing literature, studies, and systems that are significant for the development of SYNTIX. It includes a study of various technological advances in automated tabulation systems, sports management systems, and web applications in both national and international arenas. The study includes various aspects of integrating Progressive Web App features such as offline access and real-time data synchronization in overcoming the challenges of manual scoring systems and internet connectivity in sports management systems. The relevant studies provide a foundation for the proposed system in this chapter.

2.1 Related Literature and Systems

This section includes a variety of local and international studies, systems, and technical literature that relate to the objectives of SYNTIX. The gathered information provides an idea about the current trends in automated tabulation systems and sports management systems, which reflect the move towards automation from traditional systems. Moreover, this section explores the possibility of incorporating the latest web technologies such as Progressive Web Apps and their ability to provide reliability for the system by working offline and synchronizing the data in real-time..

2.1.1 The Evolution of Event Tabulation: From Manual to Digital Systems

The traditional methods of tabulation in sporting events tend to be highly prone to human error, substantial delays in declaring results, and lack transparency. As per Cualbar (2024), the way forward in solving such administrative problems lies in the automation aspect, with the use of electronic scoring systems allowing the event organizers more accuracy and efficiency [5]. Such is corroborated by the study of Ramizares (2024) [19] which brings out the importance of using web-based adaptive systems that will decrease human error in entering data into computer systems and increase the reliability of the results tabulation. Additionally, Cornelio Jr. (2025) [4]

states that in today's era, it becomes mandatory for institutions to automate their systems, especially where digitization of outputs such as the awarding of e-certificates is concerned, and Hahn (2021) [8] and Hebert (2024) [9] confirm such views in tournaments as well. In line with these studies, SYNTIX incorporates these automation principles through its real-time digital dashboard, ensuring that the tabulation for CSPC intramural events is not only efficient but also highly transparent and free from the inherent flaws of manual systems.

2.1.2 Modern Sports Management and Tournament Administration

The importance of centralized coordination in organizing sporting events in modern times cannot be underestimated; this point is confirmed by the studies conducted by Peñasales (2024) [14]. The author emphasizes that an online management system significantly improves organizational control in terms of ensuring the accuracy of records, which is hard to achieve within the framework of decentralized systems based on manual operations. The importance of automation in organizing sporting events is also emphasized in the studies of Anju (2025) [1]. The author presented a novel "Smart Sports Association Management System" that allows for real-time monitoring, which is extremely important for tournament organizers because they have to make decisions quickly under pressure from competitive sports.

In other fields, according to Patrício (2023) [13], the integrity of the scoring mechanism is greatly dependent on the judging procedure and the evaluation pattern. In addition, the technical role of data synchronization for the accurate tracking of games was pointed out by Wang (2025) [20]. In a more localized environment, Fidelino (2024) [6] stated the significance of digitalizing school organizations' events through applications for digital maturity. All these sources prove the technical relevance of SYNTIX by providing solutions for centralized management and data synchronization that can cater to the requirements of the administrative process and the need for digital maturity for the modernization of the CSPC Intramurals.

2.1.3 Progressive Web Apps (PWA) and Technical Architecture

The selected technical platform plays a key role in determining the degree of user participation and accessibility of the system in question. As reported by Pisipati and Oberai (2025), the Progressive Web Application (PWA) model represents one of the best approaches to cross-platform functionality, successfully combining strengths of both mobile and web-based applications into one coherent model [18]. Flexibility is an essential feature for a platform in a sport environment, as pointed out by Anti (2022), who indicated that access via mobile platforms was an important element of communication between participants and increased participation rates [2].

However, the liveness of the digital platform and its interaction with the audience using the digital artifacts are equally crucial for determining the effectiveness of engagement techniques employed at the event, as per Gomes (2023) [7]. This interactivity makes sure that the stakeholders do not remain merely passive spectators, but are instead actively participating in the presentation of the information. Following this idea, Permana (2023) advocated the need for mobile-based match charts and relative scorekeeping systems in order to ensure the continued adherence of the technical architecture of the application to its user-centricity principle [15]. By integrating these concepts, SYNTIX offers an offline-first PWA interface that utilizes service workers for data caching, ensuring that CSPC students can engage with the system in real time even during periods of unstable connectivity, thereby maintaining the integrity of the event's digital flow.

2.1.4 Achieving Offline Resilience and Data Synchronization

Some of the main challenges that the web application is likely to encounter, particularly in an ever-changing campus environment, include unreliable connectivity in large scale events. In this case, one of the solutions highlighted by Idowu (2022), involves the use of service workers (SW) [10]. The use of service workers enables web apps to work offline due to their ability to intercept and handle network requests behind the scenes. In this case, the service worker technology functions as a programmable network proxy that allows the web application to deliver any requested data cached on its server almost instantly regardless of the state of the local network.

The offline-first approach also finds support from the research work of Malanin (2025), which suggests that IndexedDB could be employed as a means of storage on the client side [11]. The use of IndexedDB facilitates the handling of massive quantities of structured information and, therefore, enables the program to continue working seamlessly even after losing connection. When the connection is finally reestablished, synchronization is done seamlessly to prevent any data loss or redundancy during the hectic moments of tournament scoring. SYNTIX follows these technical standards by employing both Service Workers and IndexedDB to ensure that tabulation and verification at CSPC Intramurals proceed seamlessly without data loss.

2.1.5 Software Quality Standards and Performance Evaluation

In order for a digital application to qualify as a credible and user-friendly application, it is critical that the underlying technical infrastructure conforms to stringent quality and performance criteria. As Centeno and Josephine argue in their study of contemporary software in 2024 [3], the use of up-to-date algorithms helps to ensure the reliability of automated assessment systems by eliminating the role of the human factor in data processing, ensuring that no matter how complicated the input may be, it does not interfere with the accuracy of the results obtained. On the other hand, a user-focused approach is equally important, as Permana claims that usability and visibility of scoring applications need to be validated through a series of tests, such as the System Usability Scale (SUS).

In the context of the Philippines, Pieloor (2025) [17] underlined the increasing significance of centralized systems as the basis of digital transformation in event management to ensure smooth data transmission at all organizational levels. This approach is especially applicable for educational institutions in the Bicol area, where Maligat Jr. (2020) [12] pointed out that the adoption of web-based solutions is critical in order to keep up with the times and allow smooth information transfer between all departments of universities. Based on these criteria, SYNTIX implements the centralization of an algorithmic solution, which is subject to rigorous usability testing, thus guaranteeing that CSPC Intramurals become digitally mature and competitive through open tabulation.

2.1.6 Backend Framework and Relational Database Reliability

The structural robustness of a web application depends heavily on the backend framework utilized as well as the database optimization techniques implemented. As discussed by Alfat (2015) [21], the deployment of the Laravel framework creates a well-structured software development platform due to the MVC architecture, which is indispensable for the proper handling of the data generated. In the case of SYNTIX, the framework allows for efficient segregation of complicated calculations and processes involved in scorekeeping while keeping the user interface uncomplicated and reliable in terms of generating multi-sport scores accurately. The assertion is further corroborated by the findings of Arfano (2025) [22], stating that the contemporary development ecosystem and security tools embedded in the Laravel framework are indispensable in safeguarding system data from any form of infiltration.

In order to ensure the persistence of data, SYNTIX makes use of MySQL. According to Kumar (2024) [23], MySQL provides efficient performance and stability within web applications. Also, Prasanth and France (2025) [24] note that optimizing MySQL databases at all times results in efficient monitoring and scalability. Through adherence to such database standards, SYNTIX is assured that, despite the high traffic of activities such as track and field games during intramurals, the database will be able to provide instant feedback to the leaderboard.

2.2 Synthesis of the State-of-the-Art

The literature and systems reviewed collectively underscore the importance of a major shift from traditional and manual tabulation systems to automated and web-based sports management systems. Although the local literature by Cualbar et al. and Ramizares and Ramizares has established the need for the use of electronic scoring systems to eliminate errors and ensure transparency in the process, these systems may still need a constant connection to the Internet to effectively perform their functions. On the international scene, the study by Anju 2025) and Wang (2025) underscore the potential of real-time data synchronization and smart tournament tracking systems, but these high-end systems may not be fully optimized for systems with variable connectivity like the local campus environment of Camarines Sur Polytechnic Colleges (CSPC). Moreover, though Pisipati and Oberai (2025) and Malanin (2025) offer a technical guide to "offline-first" applications with Service Workers and IndexedDB, there is a glaring absence of applying this particular Progressive Web App (PWA) technology to a cohesive sports tabulation system with multiple events in the Philippines, where most existing solutions are either web-based (requiring internet connectivity) or offline (lacking real-time synchronism).

And this is where SYNTIX claims its "State-of-the-Art" title. It does this by providing a solution that combines the reliability of offline-first storage with the performance of real-time synchronization. Unlike the systems proposed in the works of Peñasales et al. and Fidelino et al., SYNTIX has been designed to overcome the problem of "intermittent connectivity." By giving judges the opportunity to log scores even when the Wi-Fi connection is lost and then automatically syncing this information when the Wi-Fi returns, the proposed system ensures that the tournament process remains continuous. This integration of reliability and performance represents a significant improvement over existing local tabulation methods, providing a solution that can be considered "disruption-proof."

In addition, the reliability of modern web applications depends on efficient frameworks and databases. The MVC architecture and security offered by the Laravel framework, as pointed out by Alfat et al. (2015) [21] and Arfano et al. (2025) [22], is an efficient alternative to conventional scripting techniques, which are vital for safeguarding confidential student information. Similarly, the utilization of MySQL is justified by Arul Kumar et al. (2024) [23] and Prasanth and France (2025) [24], where they stress that MySQL has the capacity to scale up and monitor in real-time, which is essential for high-risk situations. Although these technologies are extensively employed in the general domain of software development, their incorporation in a sports scoring application for a local college setting is still a unique technical challenge that should be taken care of.

2.3 Gap Bridged by the Study

A large gap is seen between the theoretical benefits of digital tabulation and the actual dependability of a system within a setting of unstable network infrastructure, such as a local campus. Although current research by Cualbar and Ramizares, Ramizares focuses on accuracy of electronic or digital scoring, it must be noted that systems depend on a continuous and uninterrupted internet connection. In the absence of such a connection, the digital process is interrupted, and a return to a paper based method is required. This brings to light a major deficiency of “offline resilience” within current local solutions, making them not dependable.

This gap is bridged by SYNTIX by incorporating Progressive Web App (PWA) technology to develop a 'fail-safe' hybrid platform that is both fault-tolerant and real-time. This is achieved by using Service Workers and IndexedDB, as discussed by Idowu et al. (2022) and Malanin (2025), to ensure continuous scoring without interruption despite the presence or absence of a Wi-Fi connection. This study also bridges the gap by incorporating a synchronization sub-layer to ensure that stored data is automatically synced to the database once an internet connection is available. This ensures a reliable solution to maintain the integrity of sports management despite the technological challenges faced by institutions in Camarines Sur Polytechnic Colleges (CSPC).

Notes

[1] M. I. Anju, K. Kishore, and M. S. S. Swaroopa. 2025. Smart Sports Association Management System with Real-Time Tournament. *Cognitive Systems and Information Processing*. Retrieved March 24, 2026 from https://books.google.com

[2] E. Anti. 2022. Addressing Loneliness and Isolation with a Mobile Application: Connecting People through Sports and Shared Interests. *Theseus*. Retrieved March 24, 2026 from https://theseus.fi

[3] C. J. Centeno and R. Josephine. 2024. Automated Faculty Evaluation and Ranking System: Utilizing OCR NER and Decision Tree for a Web-Based Evaluation System. *Journal of Electrical Systems*. Retrieved March 24, 2026 from https://researchgate.net

[4] Cornelio Jr, B. L., S. Borcelo, R., & D. Siason Jr, N. (2025). Multiple Events and Multiple Users Tabulation System with E-Certificate Generation . *International Journal of Advanced Science Computing and Engineering*, *7*(1), 29–35. https://doi.org/10.62527/ijasce.7.1.240

[5] J. L. Cualbar, A. D. Candia Jr., J. M. L. Abcede, and A. R. A. Genalazo. 2024. E-Tabulation System for Events and Activities with LCD Screen Integration. *ResearchGate*. Retrieved March 24, 2026 from https://researchgate.net

[6] A. C. Fidelino , "Transforming School Organization Event Management With Microsoft Power Apps and Power Automate," *2024 9th International Conference on Information Technology and Digital Applications (ICITDA)*, Nilai, Negeri Sembilan, Malaysia, 2024, pp. 1-8, doi: 10.1109/ICITDA64560.2024.10809439.

[7] G. Gomes, T. Conte, T. Castro, and B. Gadelha. 2023. Engagement, Participation, and Liveness: Understanding Audience Interaction in Technology-Based Events. *ICEIS*. Retrieved March 24, 2026 from https://pdfs.semanticscholar.org

[8] M. G. Hahn, S. M. B. Navarro, L. De La Fuente Valentín and D. Burgos, "A Systematic Review of the Effects of Automatic Scoring and Automatic Feedback in Educational Settings," in IEEE Access, vol. 9, pp. 108190-108198, 2021, doi: 10.1109/ACCESS.2021.3100890.

[9] B. Hebert. 2024. A Web-Based Tabulation System for American Parliamentary Debate Association (APDA) Tournament. *Erik Demaine*. Retrieved March 24, 2026 from https://erikdemaine.org

[10] S. A. Idowu, A. O. Adebayo, and C. Ajaegbu. 2022. Service Workers (SW): Facilitating Offline Accessibility in Web Applications. *Journal of Advanced Research in Computing and Applications*. Retrieved March 24, 2026 from https://search.ebscohost.com

[11] V. Malanin. 2025. Implementing offline-first web apps for remote healthcare monitoring. *ResearchGate*. Retrieved March 24, 2026 from https://researchgate.net

[12] D. E. Maligat Jr., J. O. Torio, and R. T. Bigueras. 2020. Web-based knowledge management system for Camarines Norte State College. *IOP Conference Series*. Retrieved March 24, 2026 from https://iopscience.iop.org

[13] J. P. P. Patrício. 2023. Web-Based Sports Scoring System for Trampoline Gymnastics Competitions. *ProQuest*. Retrieved March 24, 2026 from https://search.proquest.com

[14] C. R. R. Peñasales, K. T. Soberano, and R. O. Algara. 2024. E-Board Sports Management Information System with SMS Support: Usability, Maintainability, Accuracy. *Academia.edu*. Retrieved March 24, 2026 from https://academia.edu

[15] Permana, P. A. G., Dewanti, P., Triandini, E., & Rusli, M. (2023, October). Developing Mobile-Based Sports Match Charts and Scoring for an Effective Match Analysis. In *2023 5th International Conference on Cybernetics and Intelligent System (ICORIS)* (pp. 1-6). 10.1109/ICORIS60118.2023.10352196

[16] P. A. G. Permana and P. Dewanti. 2024. Analysis of sports match charts and scoring applications based on website and mobile using the system usability scale method. *IEEE Xplore*. Retrieved March 24, 2026 from 10.1109/ICORIS63540.2024.10903865

[17] J. J. Pieloor, D. C. Parcasio, and M. L. Flores. 2025. Centralized Event Management Platform for Digital Transformation in Continuing Education. *Cognizance Journal*. Retrieved March 24, 2026 from https://cognizancejournal.com

[18] S. Pisipati and K. Oberai. 2025. Offline and Cross-Platform Location-based services for Healthcare: A Progressive Web Application (PWA) Approach. *ISPRS Annals*. Retrieved March 24, 2026 from https://isprs-annals.copernicus.org

[19] M. L. Ramizares and M. L. Ramizares. 2024. Development of A Web-Based Adaptive Multi-Event Tabulation System. *ResearchGate*. Retrieved March 24, 2026 from https://researchgate.net

[20] Z. Z. Wang, X. Xia, and Q. Chen. 2025. Multi-level data fusion enables collaborative dynamics analysis in team sports using wearable sensor networks. *Scientific Reports (Nature)*. Retrieved March 24, 2026 from https://nature.com

[21] L. Alfat, A. Triwiyatno and R. R. Isnanto. 2015. Sentinel web: Implementation of Laravel framework in web based temperature and humidity monitoring system. *2nd International Conference on Information Technology, Computer, and Electrical Engineering (ICITACEE)*. doi: 10.1109/ICITACEE.2015.7437768.

[22] Arfano, et al. 2025. Web Development Using Laravel Framework. *International Journal of Research in Computer Science*. doi: 10.46336/ijrcs.v6i4.1103.

[23] Arul Kumar, et al. 2024. Performance analysis of MySQL database in web applications. *OAR Journal of Engineering and Technology*. doi: 10.53022/oarjet.2024.7.1.0042.

[24] R. Prasanth and K. France. 2025. Real Time Monitoring and Continuous Optimization of MySQL Databases. *2025 6th International Conference on Data Intelligence and Cognitive Informatics (ICDICI)*. doi: 10.1109/ICDICI66477.2025.11134974.

CHAPTER 3
TECHNICAL BACKGROUND

This chapter provides a comprehensive and in-depth discussion of the research methodology and technical processes involved in the development of SYNTIX: Automated Intramurals Multi-Event Scoring and Tallying System. This chapter will also discuss the research design and system architecture framework and the specific software and hardware requirements that are essential for the successful implementation of the system. In this chapter, the systematic operational flow of the system, from the manual entry of scores by tabulators to the real-time dissemination of leaderboards using QR code, will be presented as a technical blueprint for the accuracy and transparency of the scoring process for the intramurals of Camarines Sur Polytechnic Colleges (CSPC).

3.1 Research Design and Methods

The Applied Research design was applied by the researchers, which is a research strategy that emphasizes practical applications towards solving particular issues faced by organizations or communities. The reason why the researchers used this research design in relation to SYNTIX is because there were actual issues being encountered with respect to manual scorekeeping, calculation errors, and delayed dissemination of results within CSPC.

First, the study makes use of the Software Engineering approach, which integrates qualitative data collection with the process of progressive development of the technology. The methodology used in the study is designed in such a way that the institutional regulations of the SIKLAB 2025 initiative will be turned into a real Progressive Web Application (PWA) that can enhance the management of sports and cultural activities at the college.

In order to ensure systematic development, the developers used the Agile Development Model, which is explained in depth in the next section. The model enables the developers to split the complicated requirements of the multievent system into smaller components and test each module systematically before implementing the entire system.

3.1.2 Research Instruments

The researchers designed and used specific instruments in the systematic collection of the required data for the development of SYNTIX. The specific instruments were designed to obtain the technical specifications required by the software as well as the institutional regulations in the college.

Interview Guide

The researchers prepared a semi-structured interview guide that was used in gathering qualitative data from the Supreme Student Council (SSC) Officers and official tabulators. This research instrument was used in identifying the issues that are being encountered in the manual counting process, such as errors in computations and delays in disseminating the results. This was used as a primary research instrument in determining the user requirements and interface functionalities of the system.

Document Analysis Checklist

The document analysis checklist was used to analyze the approved 2025 CSPC Intramurals Proposal (SIKLAB 2025). This instrument played an important role in extracting the "business logic" of the institution. This includes the official points distribution, list of events, and ranking criteria. This checklist ensured that every rule that is inputted in SYNTIX is verified based on the records of the college.

System Requirements Checklist

To determine the hardware and software specifications, the researchers utilized a technical checklist. This instrument was used to assess the compatibility of the Progressive Web App (PWA) with the existing devices of the tabulators and the accessibility of QR code scanning for the student body. The data gathered from this instrument provided the technical parameters for the system's architecture.

3.1.3 Data Gathering Procedure

The researchers adopted a scientific method for ensuring the validity and reliability of data collected for the development of SYNTIX. The method has been divided into various stages as follows:

Phase 1: Preliminary Preparation and Ethical Clearance

The researchers started by seeking permission to conduct the research. A formal letter of request was sent to the pressident of the Supreme Student Council (SSC) Officers of Camarines Sur Polytechnic Colleges (CSPC). The permission ensured that the researchers were able to obtain official records from the institution and conduct an interview with key stakeholders.

Phase 2: Document Acquisition and Analysis

Upon approval, the researchers obtained a copy of the Approved 2025 CSPC Intramurals Proposal (SIKLAB 2025). Using a document analysis checklist, the researchers systematically reviewed the proposal to extract the "business logic," including the scoring rules, list of events, and ranking criteria. This provided the foundational rules for the automated tallying feature of the system.

Phase 3: Conduct of Interviews and Technical Assessment

The interview process was semi-structured and conducted with the SSC Officers and tabulation officers. During the interviews, the interview guide was used to identify the challenges facing the manual process as well as the requirements of the interface. Meanwhile, the technical checklist was employed to determine the compatibility of the PWA with the existing technology of the college committees.

Phase 4: Data Integration and Validation

Finally, all gathered qualitative and technical data were compiled and analyzed. The researchers cross-referenced the interview results with the SIKLAB 2025 guidelines to ensure that the functional requirements of SYNTIX were perfectly aligned with both user needs and institutional standards. This consolidated data served as the primary input for the system's design and development phase.

3.1.4 Data Analysis Procedure

The researchers used systematic techniques to analyze the gathered data and transform them into functional requirements for SYNTIX:

- **Thematic Analysis -** Qualitative data from interviews with SSC Officers and tabulators were categorized into themes such as "Manual Calculation Errors" and "Transmission Delays" to define the system's features.
- **Rule Extraction** - The SIKLAB 2025 proposal was analyzed to extract the official points distribution and ranking criteria, which served as the basis for the system's automated scoring algorithms.
- **Technical Mapping** - Data from the technical checklist were analyzed to establish the minimum hardware and software requirements for the Progressive Web App (PWA) and QR code functionality.
- **Data Triangulation** - The researchers cross-referenced the user needs with institutional standards to ensure that the system's logic is both accurate and compliant

3.2 Theoretical or Conceptual Background

The development of SYNTIX is grounded in the principles of Information Automation and Real-time Data Processing to transform the manual scoring process of Camarines Sur Polytechnic Colleges (CSPC) into a digitalized system.

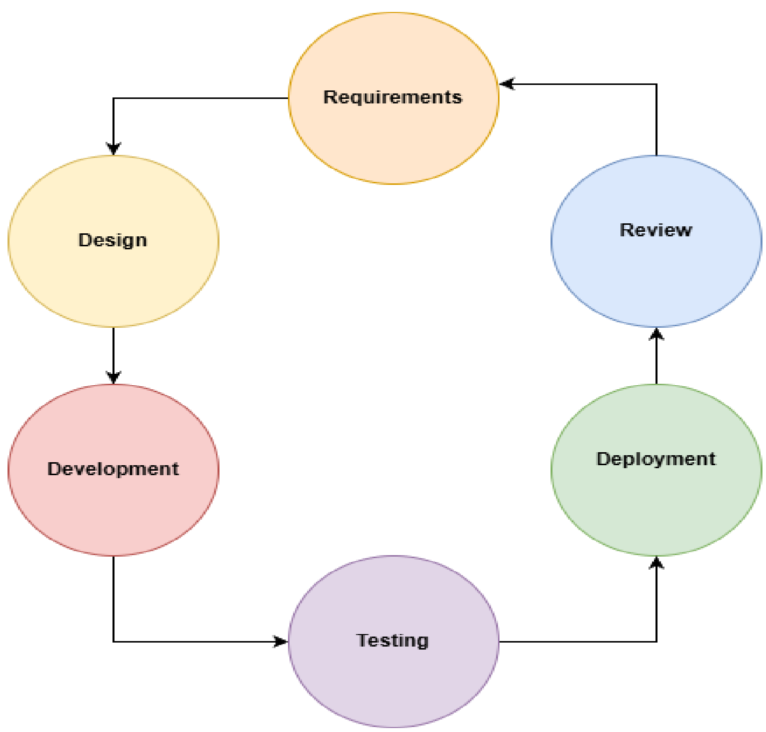
- **Automation Theory** - This study applies automation to minimize human intervention in calculating sports and cultural points. By translating SIKLAB 2025 rules into algorithms, the system reduces manual computation errors and delays.
- **Cloud-based Integration and Synchronization** - The project utilizes cloud technology for immediate data dissemination. Scores entered by tabulators are

synchronized instantly to a centralized leaderboard, ensuring transparency for the SSC Officers and the student body.

- **Progressive Web App (PWA) Framework -** The system is conceptualized as a PWA to ensure cross-platform accessibility and offline capability. This allows tabulators to input data even during intermittent Wi-Fi connection, with the system automatically syncing once the connection is restored.
- **User-Centric Design -** The framework focuses on the efficiency of the user interface for SSC Officers and tabulators. Through QR code technology, the system provides a seamless and secure method for accessing and entering event results

3.2.1 Software Development Methodology

The researchers will adopt the Agile Methodology for the development of SYNTIX. This iterative approach allows for continuous testing and integration of feedback from the stakeholders, ensuring that the system remains flexible to the requirements of the institution.



As shown in the figure above, the iterative phases of the Agile methodology used in the creation of SYNTIX can be explained. The first phase is requirements gathering, where the researchers analyze the rules and regulations set in SIKLAB 2025 as well as conduct interviews with the SSC officers and the tabulators to understand the functional requirements of the system.

The design and development processes involve the construction of the user interface and creation of the cloud-based database using the Firebase platform. The testing and deployment phase involve verifying the accuracy of the automatic counting system as well as the efficiency of the QR code reading process before presentation to the stakeholders.

This process ends with the Review stage, which involves receiving feedback from the Capstone Adviser and from users to ensure that any modifications needed will be included in the subsequent "sprint." This makes sure that the system remains adaptive and precise for use in the CSPC intramurals.

3.2.2 Design Patterns Algorithms

SYNTIX has been developed in accordance with the MVC pattern of architectural design, which is fully supported by the Laravel platform. With this pattern, it can be guaranteed that there will be a separation between the functionalities of the database and the user interface in an organized manner. In this design pattern, the Model component is responsible for handling the logic of data using MySQL, making sure that the scoring criteria are adhered to within the database. The View is used to develop the user interface, while the Controller controls the interactions between the two.

In order to achieve flawless automation and responsiveness, the use of Asynchronous JavaScript and XML (AJAX) and WebSockets (Laravel Reverb or Pusher) is incorporated in the app. Unlike other conventional systems that demand refreshing, the logic here allows the software to relay any changes done by judges and tabulators in a real time manner to the Public Dashboard, making sure that everyone gets their results instantly since the system updates its MySQL database accordingly. Besides, the algorithm of sorting is adjustable according to various sports and cultural differences, which is shown below.

Table 1: Algorithm Application and Scoring Logic

Event Category	Scoring Unit	Algorithm Used	Ranking Logic
Ball Games	Points	Tournament Bracket	Winner Advances Per Match
Track and Swiming	Time (mm:ss:ms)	Ascending Sort	Lowest Time is Rank1
Fields (Jumps, Throws	Distance (m)	Descending Sort	Highest Distance is Rank1
Cultural Events	Percentage (%)	Weighted Mean	Highest Total is Rank1

As seen from Table 1 above, the process automatically determines the type of events that need the right technique of evaluation. For competitive events involving the use of several parameters for evaluation such as in the case of pageants and dancing contests, the system utilizes the Weighted Mean Algorithm. This involves multiplying the raw scores with their percentage weights to generate a final score that cannot be altered:

$$Final\ Score = \sum_{i=1}^{n} (s_i \times w_i)$$

Where:

- s_i = the raw score given by the judge for a specific criterion.
- w_i = the assigned percentage weight for that criterion.
- n = the total number of criteria for the event.

3.3 System Architecture

The architectural design of the SYNTIX system is based on the Cloud-Based Client-Server Architecture, adopting the PWA architecture style, which guarantees availability and real-time responsiveness. The architectural design enables smooth data transmission from where the judges input the data to real-time result visualizations for the public.

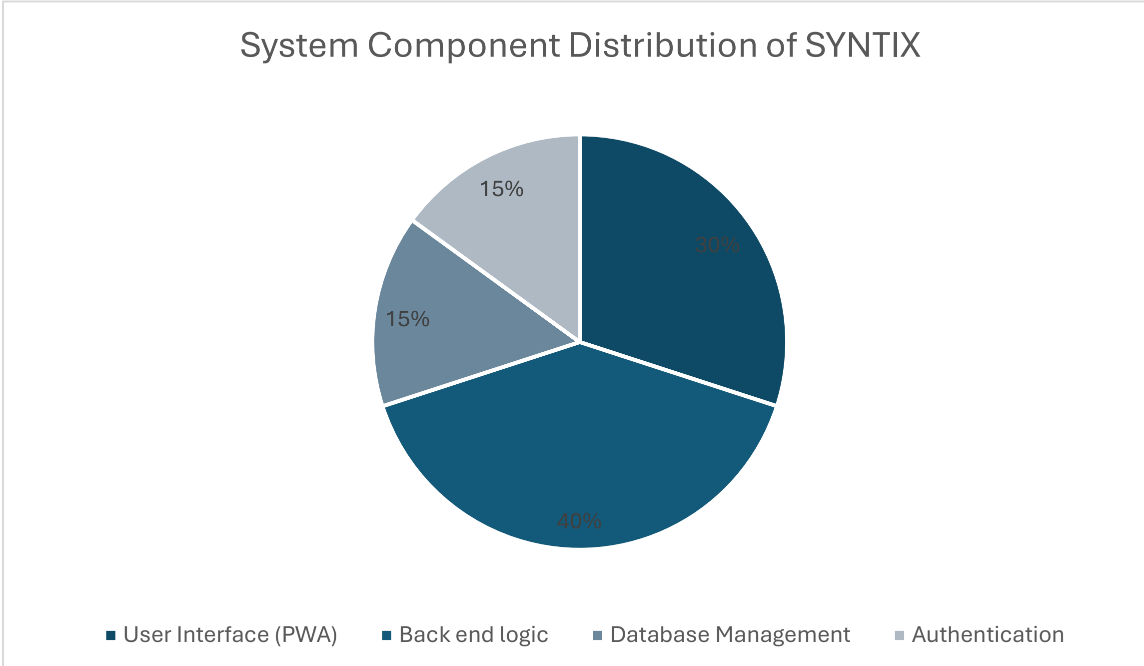


Figure 1. System Component Distribution of SYNTIX

As illustrated in Figure 1, the architecture of the system is divided into four functional components that manage the overall operations:

1. **Frontend & PWA Interface (30%):** This layer handles the user experience for both the judges (score entry) and the student body (public leaderboard) through a Progressive Web App (PWA) framework, ensuring cross-platform accessibility and offline capabilities.
2. **Backend Logic & Algorithms (40%):** This serves as the "brain" of the system, powered by the Laravel framework. It handles the execution of the Weighted Mean algorithms and dynamic sorting logic to ensure accurate and automated result tabulation.
3. **Database Management(15%):** Utilizing MySQL, this component acts as a centralized repository that ensures all event data and student profiles are stored securely with high data integrity and structured relationship mapping.
4. **Authenticatiom (15%):** Managed through **Laravel Breeze**, this component handles secure user access and role-based permissions (RBAC) to ensure that only authorized personnel can enter or modify scores.

3.4 Software Development Stack

This section provides an overview of the software tools, frameworks, and platforms used in building and running SYNTIX. These specifications ensure that the system operates efficiently, maintains data integrity through a relational database, and remains scalable for future institutional upgrades.

Table 2 presents the complete set of software specifications required for developing and deploying SYNTIX:

Table 2. Software Development Specification

Software Tool	Specification
Operating System	Windows 11 / Windows 10
Backend-Framework	Laravel (PHP)
Frontend-Languages	HTML5, CSS3, JavaScript, Bootstrap5
Database and server	Mysql with phpMyAdmin, XAMPP
Development Tools	Visual Studio Code/ PHP 8/ Laravel 12
Version Control	Git / GitHub
Browser Support	Google Chrome, Firefox, Edge, Brave
PWA Features	Service Workers, Manifest.json (for Mobile Installation)

Table 2 outlines the technical stack used to develop and deploy SYNTIX. The back-end was built using PHP 8 and Laravel 12 within the Visual Studio Code IDE to ensure robust data processing, secure session management, and the efficient execution of scoring algorithms. The front-end utilized HTML5, CSS3, Bootstrap 5, and JavaScript, integrated with Progressive Web

App (PWA) technologies, to create a responsive interface that allows the system to be installed and accessed seamlessly on both desktop and mobile devices.

For data management and testing, the system employed MySQL via phpMyAdmin and was hosted locally during development using XAMPP. The platform is designed for cross-platform compatibility on Windows, Android, and iOS, supports all modern web browsers, and leverages Git and GitHub for streamlined version control and developer collaboration.

3.5 Hardware Development Stack

The hardware development stack comprises the physical components necessary for developing and executing the SYNTIX system. This guarantees that the development platform will be able to manage real-time data processing and system implementation capabilities.

Table 3. Hardware Development Specifications

HARDWARE TOOL	SPECIFICATION
Processor	AMD Ryzen 7 (Octa-core)
Memory	16 GB RAM
Storage	512 GB SSD (NVMe preferred)
Input Devices	Built-in Keyboard and Mouse (External mouse recommended for precision).
End-User Device	Smartphone/Tablet (Android or iOS) for Tabulation testing.

As can be seen from the data provided in Table 3, the hardware requirements have been optimally configured to facilitate fast and efficient development. The main computer system is powered by an AMD Ryzen 7 processor that offers multi-core capabilities essential for the simultaneous functioning of the Laravel local server, MySQL database, and development environment.

The 16 GB of RAM included is an important part of the hardware stack because it helps with efficient multitasking and proper resource management. The developers will be able to use their IDE, along with several browser-based testing tools and real-time backend processing, without any issues in their coding process.

In addition, the hardware stack goes beyond the computer to include actual mobile devices for cross-platform testing. This is crucial for ensuring that the SYNTIX PWA is responsive and provides real-time updates. The testing process ensures that the judges' UI remains stable and responsive while processing live data during the events.

3.6 Tools and Platforms

The process of developing and implementing SYNTIX includes using server-side applications and software frameworks that support a Full-stack development approach. These tools are crucial for timely updates, version tracking, and ensuring platform-independent access to the system.

Table 4. Development Package Specifications

DEVELOPMENT TOOL	SPECIFICATION
Composer	Dependency Manager for PHP/Laravel
npm (Node Package Manager)	For managing frontend assets and PWA libraries
Laravel Vite	Frontend Build Tool and Asset Bundler
MySQL Workbench / phpMyAdmin	Database Management and Optimization Tool
GitHub	Cloud Repository for Version Control
Postman	API Testing and Endpoint Verification

The development tools, As seen in Table 4, tools such as Composer and Laravel Vite act as the technological foundation for processing the system's code. Using Vite, it is possible to conduct real-time frontend testing and hot-reloading, which is essential in ensuring accurate synchronization between the judge's inputs and the public dashboard.

MySQL Workbench or phpMyAdmin serves as the primary environment where developers can observe data flow, optimize queries, and manage relational database security in real time. This local and cloud-ready setup, working together with GitHub, ensures proper documentation of the SYNTIX codebase and prepares the system for live tabulation events.

In addition, the Laravel Routing system is utilized to improve user experience, allowing the system to switch between modules smoothly. This preserves the “App-like” nature of the PWA and ensures that judges and administrators can efficiently navigate the system without unnecessary page reloads, making it ideal for high-pressure live events.

3.7 Testing Plan

The SYNTIX testing plan used a Black Box Testing approach, focusing on the system’s external behavior rather than its internal structure.

Figure 5 presents the testing plan for SYNTIX, outlining the steps and procedures followed to ensure that all system modules function correctly and meet the required standards. This systematic approach aimed to identify and resolve potential bugs early in the process, ensuring a reliable and user-friendly experience for all users.

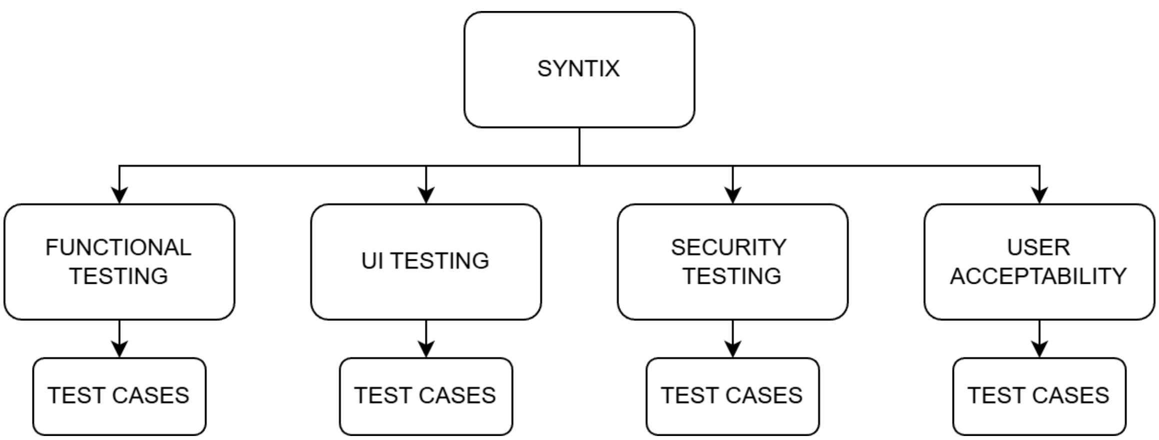


Figure 5. Testing plan

Figure 5 illustrates the framework of the SYNTIX test method. This testing approach consisted of Functional Testing, where specific Test Cases were employed to verify the key functionalities of the Admin and Judge modules. The UI Testing included the collection of UI/UX feedback to analyze the user-friendly nature of the design and conducting Responsive Testing to ensure the proper adjustment of the system across different screen sizes.

Furthermore, Security Testing ensured the protection of tabulation information by validating the authentication systems using Laravel Authentication (Breeze) and other real-time data protection strategies. This ensures that only authorized personnel can access and modify

3.7.1 Development Testing Plan

The SYNTIX system goes through several types of testing to ensure it works correctly and is user-friendly. Each type of testing has a specific purpose.

Functional Testing. This testing ensured that all features of the SYNTIX System operated correctly and met specified requirements from an end-user perspective. It primarily involved executing Test Cases that covered complete user workflows, such as an administrator setting up an event, a judge inputting scores for a specific criteria, and the subsequent automatic updating of the leaderboard. It confirmed that inputs produced the expected outputs—for instance, ensuring that weighted mean calculations were accurate—and that all application features functioned as intended.

Security Testing. This testing focused on evaluating the system's resilience against unauthorized access and ensuring consistent performance under varying loads. For the SYNTIX System, it involved rigorous testing to verify that the platform could handle multiple simultaneous judge logins during an event without crashing, checking Database Synchronization Speed to ensure quick data retrieval, and validating role-based access controls to guarantee that sensitive scoring records remained secure against unauthorized entry.

UI Testing. This testing assesses the quality of the user's visual and interactive experience with the system's interface. It involved gathering direct UI/UX Feedback to refine the design layout for better readability and ease of use, especially for judges who need to input scores quickly. Additionally, it examined the integration of the Progressive Web App (PWA) features to ensure that the navigation remained smooth and user-friendly across smartphones, tablets, and laptops.

3.7.2 System Evaluation

User Acceptability. This is the final stage of testing where the system is evaluated for overall satisfaction and operational readiness by its intended audience. It focused on confirming that the **SYNTIX** System fully met the specific needs of event organizers and tabulators, ensuring that all features were practical for live event use. Passing this stage signified the conclusion of the testing phase, validating that the system was stable, effective, and ready for full deployment.

Test Case. Test Cases are a set of conditions and variables under which a tester determines whether a system satisfies requirements and functions correctly. For the SYNTIX System, detailed test cases were essential for verifying core workflows, such as a judge successfully submitting a score, the admin approving results, and the accurate generation of the digital medal tally to ensure all logical paths functioned as intended.

Notes

[1] Agile Alliance. 2025. *The Agile Roadmap: Adaptive Systems for Academic Institutions*. Retrieved April 11, 2026 from http://agilemanifesto.org/

[2] Archibald, J. 2025. *Progressive Web Apps: Advanced Service Worker and Offline Caching Strategies*. Tech-Press, San Francisco, CA.

[3] Chen, X. and Wu, Y. 2025. Optimization of Real-time Data Synchronization in Cloud-based Scoring Systems. *Journal of Software Engineering and Applications* 18, 2 (2025), 45–60.

[4] Garcia, M. 2025. *Next-Generation UI/UX: Designing with Utility-First CSS and Tailwind v4*. Creative Code Publishing, Austin, TX.

[5] Haverbeke, M. 2025. *Eloquent JavaScript: A Modern Introduction to Programming* (5th ed.). No Starch Press, San Francisco, CA.

[6] ISO/IEC. 2025. *Software Quality Standards for Progressive Web Applications: 2025 Updates* (ISO/IEC 25010:2025).

[7] Johnson, R. and Smith, K. 2026. *Cloud-Native Architecture for Real-Time Competition Management*. Academic Tech Press, Boston, MA.

[8] Khan, M. E. 2025. Advanced Black Box Testing Strategies for Automated Tabulation Systems. *Int. J. Comput. Sci. Netw. Secur.* 25, 1 (2025), 12–28.

[9] Myers, G. J. 2025. *The Art of Software Testing: New Perspectives on Automated Verification* (4th ed.). John Wiley & Sons, Hoboken, NJ.

[10] Tailwind Labs. 2025. *Tailwind CSS v4: The Future of Utility-First Design Documentation*. Retrieved from https://tailwindcss.com/docs

[11] Otwell, T. 2026. Laravel: The Definitive Guide to Modern PHP Web Development (12th ed.). Leanpub. Retrieved from https://laravel.com/docs

[12] Oracle. 2025. MySQL 8.4 Reference Manual: High-Performance Relational Database Design. Oracle Press. Retrieved from https://dev.mysql.com/doc/

[13] Stauffer, M. 2025. Laravel: Up & Running: A Framework for Real-Time Applications (4th ed.). O'Reilly Media, Sebastopol, CA.